

Element C Worksheet

Design Requirements

Design Requirement 1 (High Priority)

- The design must be safe to use for handlers, veterinarians, and dragons.
- The solution must not injure the clients for it to be considered a success. If the solution (utilized under expected conditions) causes harm to the zookeepers, veterinarians, or dragons, it is a failure as the solution has created a dangerous situation, as the initial purpose was to improve the safety and usefulness of the current solution. The goal of this box is to improve overall safety for everybody to work with the dragons and care for them. This is a very important point for the clients as the safety of both people and the animals is a top priority for the zoo.

Design Requirement 2 (High Priority)

- The dragons must not be able to turn around in the box.
- If the dragons can turn around, then the design will be less safe for the zookeepers and the dragons during medical procedures. The clients stated that the dragons must not be able to turn around in the box because if the dragons can turn around, they are a danger to the zookeepers, as they could get bitten by the dragons. The better that the dragons are held in the box, the better it will be to take x-rays and ultrasounds as well.

Design Requirement 3 (High Priority)

- The design requires a spot for an X-ray plate (15.75"x12") beneath and to the side of the animal.
- The box needs to be able to be set to take x-rays of the dragons. The Zoo stated that they require a way to take X-rays of the dragons without removing them from the box. Currently, they have no way to x-ray the dragons without sedating them. The box must have a way for them to take X-rays of the dragons so that they can take better care of the animals. The x-ray plates that they will use for the dragons are 15.75"x12" and must be placed under and to the sides of the box. The materials of the box must not interfere with the x-ray.

Design Requirement 4 (High Priority)

- Holds 250 lbs
- The solution needs to be both strong and adaptable to suit multiple sizes of dragons from small juveniles to full-grown females, and 9' male dragons. The solution has to be simple to adjust for different sizes of dragons while retaining the strength for larger, heavier dragons. The box will cause an incredibly dangerous situation if it fails to hold the weight.

of full-grown Komodo dragons. The dragons are incredibly dangerous with their bite and wiping of their tails, therefore the box must be able to easily withstand these forces.

Design Requirement 5 (High Priority)

- The solution must have the ability to take an Ultrasound of the animals.
- The solution has to have space around the female dragons such that the veterinarians are able to take an ultrasound of the dragon while it remains in the box. The zoo needs to be able to see if the females have eggs and need a way to safely check for the eggs. The solution must have a slot roughly 6"x18" so that the probe (2.5"x4") easily fits in and has space to move around to get the right angle on the dragons.

Design Requirement 6 (High Priority)

- The box needs a window to view the dragons through.
- The zoo raised interest in having a window so that they can see how the dragons are oriented when they need to reach into the box to work with the dragons. The veterinarians must be able to see the orientation of the head of the dragons so that they are not in danger of getting bitten by a dragon when they reach the box. They also need a window for safety while letting the dragons back out of the box.

Design Requirement 7 (High Priority)

- There must be a way to draw blood from the dragons while they are in the box.
- The vet needs to be able to draw blood from the tails of the dragons to run tests on them. The tails of the dragons should be folded into the box, which will keep the tails from moving around too much and aid with the ability to draw blood. The veterinarians will need a door through which they can reach to draw blood.

Design Requirement 8 (Low Priority)

- The box is easy to clean out between uses.
- Talking with some of the zookeepers informed us that the dragons can be messy. They stated that they would need the box to be made of a material that can be easily washed off and disinfected. When interviewing the head of the reptiles at the zoo, he suggested a material, polypropylene, which is a hard plastic that could be easily cleaned and disinfected between uses.

Design Requirement 9 (Low Priority)

- The solution would ideally have the ability for zookeepers to clip the Komodo Dragons' claws while the dragon remains in the box.
- The zookeepers have to clip the dragon's claws from time to time, and it would be good for them if they were able to do it while the dragons are in the box. This was mentioned, however, less important than all the other points that were talked about. One problem that was mentioned for this was the different dragons being different sizes, and the positions

that the dragons could be in while in the box, causing problems for trying to have places to clip their claws.

Criteria from Door Design Matrix

Focusing on medical imaging, our group decided to make a decision matrix for the side door component of the windows. The zookeepers requested a special opening for performing ultrasounds on their female Komodo dragons. Our criteria surrounding this challenge were chosen with the goal of safe and efficient utilization to ensure long-lasting use of the feature.

Simplicity of latching/opening - 0.15

Latching ability - 0.15

X-ray Compatibility - 0.25

Range of operator motion - 0.2 1/13

Durability - 0.15

Reliability - 0.1

Sum (should = 1) - 1

The criteria we chose for the ranking of our design choices circulated mainly around the importance of the security of the lock, the effectiveness of the feature, and the longevity of our design.

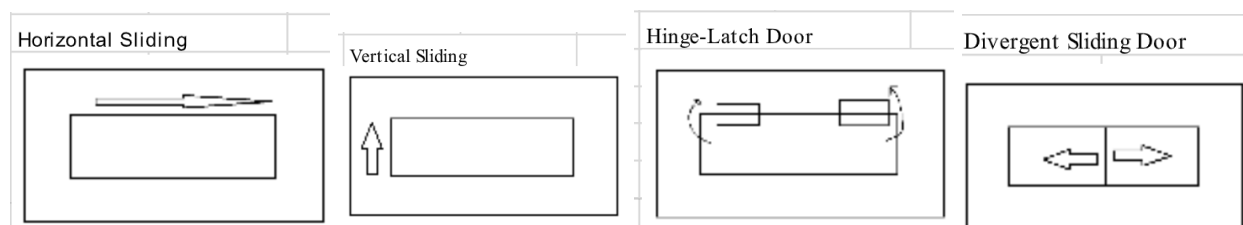
“Latching ability” and “Simplicity of latching/opening” refer to the security of the lock. With continued use of this feature during check-ups for dragons other than the resident female dragons, it is of high importance to ensure the safety of the lock on the opening. While this feature is helpful for dragons that need an ultrasound, the same box may be utilized to transport dragons that do not require this exam. For those cases, the lock must remain secure at all times to guarantee the safety of veterinary technicians. If the lock were to fail, any zoo employee around the box would be put in danger of attacks made by enclosed dragons as this would allow for the claws or tail to be exposed. In the prevention of this event, the simplicity of the lock should be prioritized to streamline the process of closing and locking the opening. Additionally, the “latching ability” or the ability to lock and stay locked should be prioritized to avoid any unnecessary injury.

“X-ray compatibility” and “Range of operator motion” are integral aspects of the effectiveness of the feature. When tasked with this project, the zookeepers emphasized the importance of the box to be able to be X-rayed and to be easy to use in a variety of orientations. One of the main purposes of the box is to be used in medical imaging, primarily of X-ray exams. Fulfilling these criteria is a major goal of the box and must be prioritized by being given a larger proportion in the design matrix.

“Durability” and “Reliability” are significant components of the goal for longevity of our design. While we prioritize the effectiveness of the box for its main purpose of transporting and storing the dragon safely and effectively, the ability of the box to maintain its strength and quality over an extended period is additionally important for the use of the box. In the design matrix, these qualities are given a relatively smaller proportion because they are not necessarily integral to the main workings of the box, but they are important aspects of a useful design even after our interaction with it.

Specifically, “Durability” refers to the strength of the box and the box’s resilience to expected and unexpected forces. Given the potential dangers posed by exposure to unrestrained komodo dragons, users of the box may need to unexpectedly shut a side door quickly or with a greater amount of force than anticipated under usual use cases. As such, durability involves (but is not limited to) the ability to maintain user safety in both expected and unexpected settings.

Design Choices



Our design choices are the result of group brainstorming. Each member of the medical imaging group thought of one or two ideas to bring to the table. Once we discussed our ideas and came to a common consensus on what each choice entailed, rudimentary diagrams were made and we began independent comparisons of the designs to our criteria (explained above). Once design matrices were completed, we averaged the outcomes to come to a mathematically achieved agreement on the comparison of our design choices to each other.

X-Ray Plate Holder

Veterinarians at the Nashville Zoo expressed a need to image the Komodo dragons both laterally and from above (vertically). For a crate-shaped design, enabling both lateral and vertical imaging would require the ability to perform an X-ray with the X-ray detector plane facing the sides of the crate and the top of the crate.

An issue arises when X-raying